

Enabling Asymmetric Alkynylation of Azaaryl Aldehydes with Soai Autocatalyst

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Dedicated to the memory of Professor Per Ahlberg

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A new sequence of autocatalysis–catalysis in asymmetric alkynylation of azaaryl aldehydes in the presence of Soai (5-pyrimidyl)alkanol as chiral catalyst is documented. The auto-

catalyst can be generated from almost racemic environment to up to 99.5 % *ee* and subsequently propagates this chirality to provide propargylic alcohols in up to 86 % *ee*.

Introduction

Finding new methods to access chiral molecules – *asymmetric synthesis* – has in the past 40 years become a key activity for organic chemists in academic laboratories and in industry. Despite the significant progress in enantioselective transformations, focus has been on the use of enantiomerically pure chiral molecules available from the chiral pool, since a linear relationship of the *ee* values between the chiral catalyst and the product is anticipated. More recent developments show that enantiomeric enrichment may also be acquired starting from achiral or racemic precursors in absence of chiral catalysts.^[1] Thus, organic or organometallic reactants are obtained as single enantiomer crystals to be used as reactant in absolute asymmetric synthesis.^[2] Alternatively, a near equilibrium deracemization may be achieved through grinding-induced attrition.^[3]

In reality, asymmetric amplification may arise as a consequence of diastereomeric perturbations in the structure of the active complex [positive nonlinear effect, abbreviated as (+)-NLE] to yield a product of higher enantiopurity than the chiral catalyst.^[4] Several reports described its use in mechanistic studies, but this effect has not been extensively applied in asymmetric catalysis. Nevertheless, and despite the structure requisite of a rigid γ -imino aldehyde, Soai et al. remarkably established that asymmetric amplification is valuable in autocatalytic reactions as in the addition of diisopropylzinc on several azaaryl aldehydes.^[5]

However, the Soai reaction requires specific experimental conditions, i.e. a sequential procedure of 3–4 consecutive

reactions, and thus can be described as Frank-like reaction network operating in a closed system to give rise to absolute asymmetric synthesis under kinetic control.^[1]

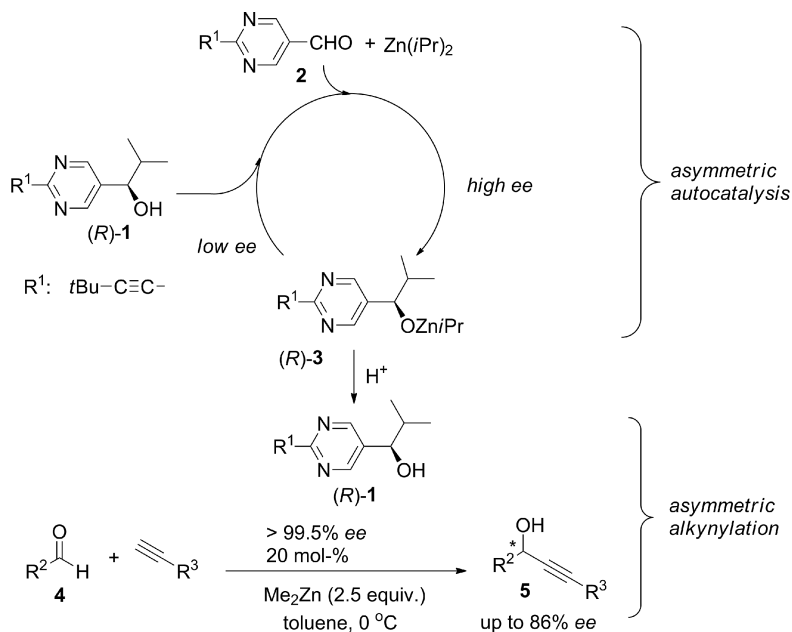
Thus, a catalytic sequence where an autocatalyst evolves with asymmetric amplification and could act as catalyst in an ensuing asymmetric reaction, would become advantageous; because high enantiopurity is generated from a minimal fraction through such a proliferation process and could potentially propagate in a range of chemical transformations (Scheme 1). Ideally, and perhaps more challenging, such a catalytic system should operate in a two-step one-pot procedure. In addition, this autocatalysis–catalysis sequence opens for alternative asymmetric processes, as it does not rely on the chiral pool. In our approach to such an autocatalytic system we made use of Soai's compound **1**.

Indeed, a practically perfect asymmetric amplification was observed in the asymmetric autocatalysis of (5-pyrimidyl)alkanol **1** by addition of $\text{Zn}(i\text{Pr})_2$ to (5-pyrimidyl)aldehyde **2**.^[6] Soai's group also showed that a consecutive system of asymmetric autocatalysis and asymmetric catalysis provided enantioenriched compounds with significantly larger amplification of *ee* in the autocatalytic alkylation of aldehydes using the formed (5-pyrimidyl)alkanol **1**.^[7] Surprisingly, this strategy has yet to be extended to a range of potentially useful asymmetric transformations other than Soai's alkylation systems. These properties indicated to us that **1** should be suitable for propagation of chirality in the alkynylation of aldehydes as a model reaction.

This reaction is particularly suitable because the zinc alkynylides reagents could be generated in situ and the versatility of the products that serve as precursors for a variety of chiral targets.^[8] In 1990, Niwa and Soai reported the first enantioselective addition of alkynylzinc reagents to aldehydes using amino alcohols as catalysts to provide propargylic alcohols in the range of 7–35 % *ee*.^[8e] The limited

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Scheme 1. Consecutive asymmetric catalyst amplification and subsequent alkylation of aldehydes.

enantioselectivity of the reaction led to the development of more efficient chiral ligands, which have successfully been applied in these reactions. Examples include BINOL and its derivatives,^[9] amino alcohols,^[10] imino alcohols,^[11] sulfonamides,^[12] oxazolidines^[13] and β -hydroxy amides.^[14] More recently, this methodology also provided access to chiral propargylamines.^[15]

Results and Discussion

Herein, we report an unprecedented enantioselective addition of alkynylzinc reagents to azaaryl aldehydes in the presence of (*R*)-**1** as chiral catalyst to generate enantio-enriched propargylic alcohols in up to 86% *ee*.^[16] In this communication, we also demonstrate the possibility and the advantage of a one-pot sequence combining autocatalysis with catalysis. To the best of our knowledge, no use of enantio-enriched autocatalyst to promote a different enantioselective reaction such as alkylation has been reported previously.

To evaluate the efficiency of autocatalyst (*R*)-**1** for asymmetric alkylation, we initially focused our efforts to establish the optimal working conditions in the reaction of 3-pyridinecarboxaldehyde (**4a**) with phenylacetylene (Table 1). (*R*)-**1** was prepared beforehand by asymmetric self-replication with > 99.5% *ee* following Soai's procedure.^[17a]

In our first attempt, the alkynylzinc reagent was prepared in situ by deprotonation of phenylacetylene with dialkylzinc in toluene. Thus, stirring both starting materials in the presence of 20 mol-% of (*R*)-**1** (> 99.5% *ee*) at room temperature for 22 hours followed by addition of aldehyde **4a** at 0 °C provided **5a** in a low yield of 22% and 67% *ee* (Table 1, entry 1).^[18] The low yield was attributed to the side reaction

Table 1. Optimisation of reaction conditions for the enantioselective addition of phenylacetylene to 3-pyridinecarbaldehyde.

Entry	ZnR ₂ ^[a]	ZnR ₂ /alkyne	Temp. [°C]	Yield [%] ^[b]	<i>ee</i> [%] ^[c,d]
1 ^[e]	Me ₂ Zn	3:4	0	22	67 (<i>S</i>)
2	Me ₂ Zn	3:4	0	64	68 (<i>S</i>)
3	Me ₂ Zn	1.4:1.4	0	11	53 (<i>S</i>)
4	Me ₂ Zn	2.5:7	0	91	76 (<i>S</i>)
5	Me ₂ Zn	2.5:7	r.t.	83	64 (<i>S</i>)
6	Me ₂ Zn	2.5:7	−40	42	62 (<i>S</i>)
7 ^[f]	Me ₂ Zn	2.5:7	0	84	56 (<i>S</i>)
8 ^[g]	Me ₂ Zn	2.5:7	0	70	74 (<i>S</i>)
9	Et ₂ Zn	2.5:7	0	54	43 (<i>S</i>)
10	<i>i</i> Pr ₂ Zn	2.5:7	0	69	42 (<i>R</i>)

[a] Reactions were carried out under nitrogen, pre-formation of the alkynylzinc species was performed at 75 °C. [b] Isolated product after column chromatography. [c] *ee* values were determined by HPLC with a Daicel Chiralcel OD column. [d] Absolute configuration was determined by analysis of the corresponding Mosher ester ¹H NMR spectrum. [e] Pre-formation of the alkynylzinc species was performed at room temp. for 22 h. [f] 10 mol-% of (*R*)-**1**. [g] 20 mol-% of DiMPEG 2000 as additive.

of the addition of Me₂Zn to the aldehyde to give the alkyl adduct instead of the corresponding propargylic alcohol **5a**. Hence, the deprotonation of phenylacetylene by dimethylzinc in toluene is a relatively slow step. Heating the alkyne and Me₂Zn at 75 °C for 2.5 h helped overcome this side reaction (Table 1, entry 2). However, decreasing the amount of Me₂Zn and phenylacetylene to an equimolar mixture gave a disappointing yield of only 11% and a decrease in *ee* values (Table 1, entry 3). In this case, ¹H NMR analysis

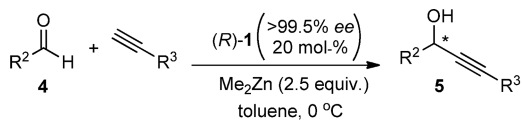
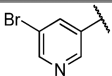
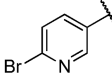
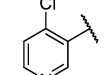
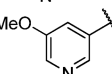
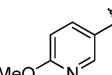
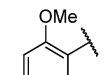
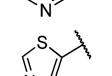
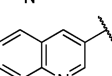
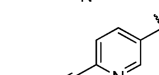
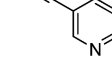
of the crude revealed only 24% conversion to the corresponding propargylic alcohol **5a**. In an attempt to push the reaction to completion without increasing the risk of methylation, we decided to increase the amount of phenylacetylene compared to Me_2Zn . Gratifyingly, a ratio of 2.5:7 in Me_2Zn /alkyne not only led to an increase in yield from 64% to 91% but also increased the enantioselectivity of the reaction from 68% to 76% *ee* (Table 1, entry 4). A temperature increase from 0 °C to room temperature or decrease from 0 °C to –40 °C all led to lower yields and *ee* (entry 5 and 6). Lowering the amount of (*R*)-**1** from 20 to 10 mol-% gave product **5a** with lower yield and *ee* (Table 1, entry 7).

Several groups reported that the use of DiMPEG 2000 [poly(ethyleneglycol) dimethyl ether, MW 2000] markedly improved the enantioselectivity in the addition of alkynes to aldehydes.^[11a,19] However, in our hands, addition of 20 mol-% DiMPEG did not show any beneficial effects on this reaction leading to nearly the same enantiomeric purity with a slight decrease in yield (Table 1, entry 8). Finally, we decided to examine other dialkylzinc reagents as precursors of zinc alkynylides, anticipating an easier deprotonation of the alkyne, although with a higher risk of competing alkylation reaction. In addition, this might induce changes in the structure of the catalytically active zinc aggregates. Indeed, the use of Et_2Zn not only resulted in lower yield (due to the formation of the alkyl adduct) but also decreased the enantioselectivity (Table 1, entry 9). Surprisingly and more interesting, the reaction conducted with the more reactive $i\text{Pr}_2\text{Zn}$ provided the opposite enantiomer of the product with 42% *ee* and good yield (Table 1, entry 10). But, again products from competing alkyl addition side reaction were observed.

Having optimized the reaction conditions (Table 1, entry 4), we decided to examine the scope of the propagation of autocatalytic chirality. We used autocatalyst (*R*)-**1** to promote the reaction of alkynylzinc with substituted nicotinaldehyde. Results depicted in Table 2 demonstrate that alkynylation proceeds with high enantioselectivity up to 86% *ee* and good yields, and no alkylation by-products were detected. While the reaction is compatible with a range of substituents on the substrate, these results also reveal a general trend for the alkynylation, e.g., substituents at the 5-position seemed to be more suitable (Table 2, entry 1 and 4). However, substituents at the 4-position had a detrimental effect on the enantioselectivity as the propargylic alcohol was obtained as a racemate (Table 2, entry 6). The observed failure could be attributed to σ -EWG at C2 and C4 resulting in lower Lewis basicity at nitrogen and thus less coordination to the active catalyst (Table 2, entries 2, 5 and 6). It is important to mention that alkynylation also works with the regioisomers, 2- and 4-pyridinecarbaldehydes, but the propargylic alcohol rearranged to the corresponding enone.^[20] The same rearrangement was observed when using ethyl propiolate as nucleophile. Another commercially available and highly versatile nucleophile TMS-alkyne was successfully employed to yield the product with high enantioselectivity (Table 2, entry 11). As we mentioned above, zinc alkynylide formation required heating of the alkyne and di-

alkylzinc in toluene at 75 °C. This is above the boiling point of TMS-alkyne and makes its use very difficult. We decided to pursue our studies with phenylacetylene as nucleophile for these practical reasons.

Table 2. Asymmetric alkynylation of azaaryl aldehydes.

					
Entry ^[a]	R ²	R ³	Prod.	Yield % ^[b]	<i>ee</i> % ^{[c][d]}
1		Ph	5b	77	86 (S)
2		Ph	5c	80	rac.
3		Ph	5d	87	27 (S)
4		Ph	5e	72	62 (S)
5		Ph	5f	57	8 (S)
6		Ph	5g	74	rac.
7		Ph	5h	66	40 (S)
8		Ph	5i	71	67 (R)
9		Ph	5j	76	65 (R)
10		Ph	5k	40	63 (S)
11		TMS	5l	65	84 (R)

[a] All the reactions were carried out in 2.6 mL of toluene under nitrogen, pre-formation of the alkynylzinc specie was performed at 75 °C, Me_2Zn (0.77 mL, 1.2 M in toluene) and alkyne (2.6 mmol), 2 h, then room temp. and ligand **1** (20 mol-%), 1 h, then aldehyde (0.37 mmol), 18–22 h, 0 °C. [b] Isolated product after column chromatography. [c] *ee* values were determined by HPLC analysis of the corresponding products. [d] Absolute configurations were determined by analysis of the corresponding Mosher ester ¹H NMR spectrum.

Interestingly, autocatalyst (*R*)-**1** was also successful with other heterocycles providing a moderate 40% *ee* in the reaction with 5-thiazolecarbaldehyde (Table 2, entry 7), whereas 3-quinolinecarbaldehyde afforded the corresponding product with increased enantioselectivity of 65% *ee* (Table 2, en-

try 8). Notably, quinolinecarbaldehyde also provided the corresponding propargylic alcohol **5i** with the same absolute configuration as the autocatalyst (*R*)-**1**. To explore the extent of the substituent effect on the stereoinduction at the 6-position on the pyridine, we first investigated the reaction of 6-(3,3-dimethylbut-1-yn-1-yl)nicotinaldehyde, a reminiscence of the Soai aldehyde. The latter was converted into alkynol **5j** in 76% yield, with 65% *ee* (Table 2, entry 9). More notable is that formation of alkynol **5j** also favored the *R* enantiomer in the presence of (*R*)-**1**. In sharp contrast, 5-(3,3-dimethylbut-1-yn-1-yl)nicotinaldehyde exhibited opposite enantioselectivity and low reactivity, providing the *S* enantiomer of **5k** in 63% *ee* (Table 2, entry 10).

To verify whether an asymmetric induction may have been promoted by alkynylation product **5**, i.e. asymmetric autocatalysis, a set of reactions was performed. First, the *ee* of product **5a** was analyzed over time as aldehyde **4a** was treated with alkynylzinc in the presence of catalyst (*R*)-**1**. These experiments revealed that product **5a** was formed at its highest 79% *ee* at the early stages of the reaction; followed by erosion to 72% *ee* at the end of the reaction, which indicated that no asymmetric amplification occurred in the alkynylation reaction.

We then performed several alkynylation reactions in the absence of the Soai catalyst but using 20 mol-% of the enantioenriched products **5a**, **5d**, and **5e**. All experiments gave the corresponding alkynols in racemic mixture after subtracting the 20 mol-% of the initial chiral product, which shows the absence of asymmetric autocatalysis of alkynols. This provides evidence that asymmetric alkynylation is catalyzed by Soai alcohol (*R*)-**1**.

Furthermore, it was appealing to examine the correlation between the *ee* of the catalyst derived from (*R*)-**1** and *ee* of product. Thus, a series of alkynylations of 3-pyridinecarbaldehyde (**4a**) with phenylacetylene catalysed by 20 mol-% of (*R*)-**1** with variable *ee* values were performed. Interestingly, the variation of *ee* values of precursor (*R*)-**1** generated asymmetric alkynylation and a positive nonlinear effect is observed (Figure 1).

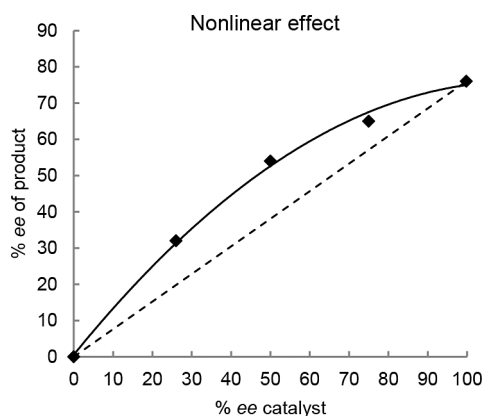
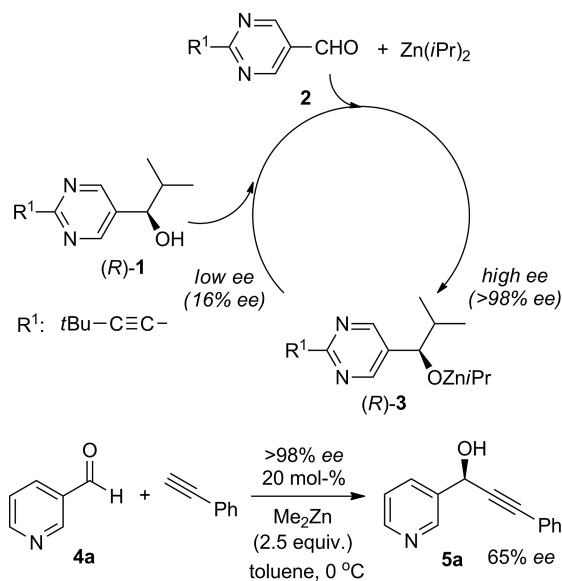


Figure 1. Plot of *ee* of propargylic alcohol **5a** as a function of the *ee* value of catalyst (*R*)-**1**. The *ee* values were determined by HPLC with a Daicel Chiralcel OD column.

This positive effect may be attributed to aggregation patterns of (*R*)-**1**/Zn to form homo- and heterochiral dimer species with different reactivity and enantioselectivity.^[4] However, as the alkynylation products are structurally similar to Soai product, such a (+)-NLE might also stem from a synergistic interaction between (*R*)-**1**/Zn and **5**/Zn. Therefore, further investigations are carried out to elucidate this question.

Due to the high efficiency of the amplification of *ee* in asymmetric autocatalysis of (*R*)-**1**, we next focused our attention on the autocatalysis–catalysis sequence without isolation of (*R*)-**3** as shown in Scheme 2. We envisioned a proliferation sequence in which (*R*)-**1** is able to amplify itself enantiomerically, and consecutively become a chiral catalyst for the alkynylation of aldehydes. To do so, 20 mol-% of (*R*)-**1** with > 98% *ee* were generated in situ through autocatalytic amplification by reaction of 2-(*tert*-butylethynyl)pyrimidine-5-carbaldehyde (**2**) with *i*Pr₂Zn in presence of an initial amount of 2 mol-% of (*R*)-**1** originally with 16% *ee*, in three consecutive cycles. Progress of asymmetric amplification was monitored by sampling and HPLC analyses and showed a prompt increase in *ee* already in the first cycle (see supporting information). Thus, the isopropylzinc alkoxide (*R*)-**3** becomes an efficient catalyst in the enantioselective alkynylation of nicotinaldehyde **4a** to yield 91% of the corresponding propargylic alcohol **5a** with 65% *ee*.



Scheme 2. Asymmetric alkynylation of nicotinaldehyde.

In summary, Soai's autocatalyst **1** – which is readily available in >99% *ee* by asymmetric amplification from virtually achiral precursor – becomes a catalyst in an ensuing enantioselective alkynylation of azaaryl aldehydes with moderate to good level of asymmetric induction (up to 86% *ee*). This observed specific reactivity confirms the uniqueness of the Soai autocatalyst. Interestingly, the reaction exhibits potential for a two-step one-pot autocatalysis–catalysis sequence as the active isopropylzinc alkoxide **3** could be amplified

enantiomerically to >98% *ee*, and used without isolation. Further investigations are currently focused to address the working mechanism in order to access a more efficient one-pot process leading to improved *ee*. This provides entry to chiral catalysts beyond the traditional chiral pool and indicates a potentially novel approach to access enantiomerically pure molecules.

Experimental Section

General Procedure for the Asymmetric Alkynylation Reaction: In an oven-dried flask under an inert atmosphere of nitrogen a solution of Me₂Zn 1.2 M in toluene (0.77 mL, 0.93 mmol) was added slowly to a solution of alkyne (2.6 mmol) in 2.6 mL of dry toluene. The resulting solution was stirred at 75 °C for 2 h. Then the reaction mixture was cooled down to room temperature and a solution of ligand **1** (0.075 mmol) in 0.3 mL of toluene was added and stirred at room temp. for 1 h. The reaction mixture was then cooled to 0 °C and the aldehyde (0.373 mmol) was added and stirring continued for 18–22 h. The reaction was quenched by adding aqueous saturated NH₄Cl (2.5 mL) and stirred vigorously for 10 min. The toluene phase was separated, and the aqueous phase was extracted with EtOAc. The combined organic fractions were dried with Na₂SO₄, filtered, and concentrated under reduced pressure. The crude product was then purified by flash column chromatography on silica gel.

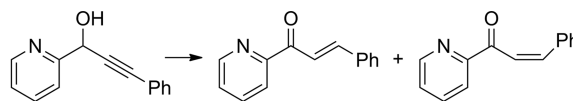
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[20] When the reaction was conducted with 2- and 4-pyridinecarbaldehyde, we observed a Raphael rearrangement of the alkynol into the corresponding enone:



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